「First degree.

Consider an equation $y' + a(t)\cdot y = 0$. Let's solve this.
If $y(t_0) = 0$ for some $t_0 \in I$, then since $\tilde{y} \equiv 0$ is a solution s.t. $\tilde{y}(t_0) = 0$, the uniqueness theorem gives $y \equiv 0$. Now, assume $y > 0$ for all $t \in I$.
Then, we obtain $\dfrac{y'}{y} = (\log y)' = -a(x)$, so

$$\log y = -\int_{t_0}^{t} a(x)\,dx \Rightarrow y = \exp\left(-\int_{t_0}^{t} a(x)\,dx\right)$$

「Non-Homogeneous Case

Consider an equation $y' + a(t)\cdot y = b(t)$.
We already know the Homogeneous Solution

$$y_H = \exp\left(-\int_{t_0}^{t} a(x)\,dx\right).$$

And, the Particular Solution is

$$y_p(t) = y_H(t)\cdot\int_{t_0}^{t} \frac{b(x)}{y_H(x)}\,dx$$

Therefore, the general Solution is

$$y(t) = y_H(t)\cdot\left[C + \int_{t_0}^{t} \frac{b(x)}{y_H(x)}\,dx\right]$$